

Faithfulness Probes for Retrieval-Augmented Summarisation

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Abstract

We present ProbeCheck, a token-level attribution probe approach to grounded summarisation. Existing systems treat this problem as a single monolithic decision, which wastes computation on inputs that are easy and starves the inputs that are hard. ProbeCheck instead allocates effort where the evidence is ambiguous. On GroundSum we reach 83.1 faithfulness F1, against 76.4 faithfulness F1 for the NLIJudge baseline, while remaining trainable on a single mid-range GPU.

1 Introduction

Work on grounded summarisation has converged on a small number of architectures that differ more in their training recipes than in their inductive biases. Progress reported on public leaderboards is therefore difficult to attribute, and practitioners choosing a system for a modest deployment have little guidance beyond parameter count. We take the opposite starting point and ask what can be achieved when the compute budget is fixed in advance at a single mid-range GPU.

Our contribution is ProbeCheck, which reorganises grounded summarisation around the observation that most inputs are decided by a small fraction of the available evidence. Rather than scaling the model, we make the decision procedure adaptive. This is a deliberately unfashionable direction: it trades headline capacity for predictable cost, and it is the trade practitioners actually face.

We evaluate on GroundSum, a collection of 15,600 examples assembled for grounded summarisation. Against the NLIJudge baseline, ProbeCheck improves faithfulness F1 from 76.4 to 83.1. An ablation shows that collapsing the probe to a sentence-level classifier costs 5.5 points of faithfulness F1, which locates the gain in the component we claim is responsible rather than in incidental tuning.

The remainder of this paper is organised as follows. Section 2 describes ProbeCheck in detail. Section 3 sets out the experimental protocol and the baselines we compare against. Section 4 reports results and ablations. Section 5 discusses the limitations of the approach, of which the most serious is that the probe is trained on extractive summaries and its calibration drifts when the generator paraphrases heavily.

2 Method

ProbeCheck has three stages: a cheap scorer that produces a provisional decision, a confidence estimator that decides whether the provisional decision is safe, and an expensive scorer that is invoked only when it is not. The three stages share an encoder, so the additional parameter cost over the NLIJudge baseline is under four percent.

The confidence estimator is the component that makes the token-level attribution probe framing pay. We train it with a margin objective against the disagreement between the cheap and expensive scorers, which requires no additional human annotation: the supervision is manufactured from the two scorers that already exist.

Training proceeds in two phases. The cheap and expensive scorers are trained jointly for the first phase, then frozen while the estimator is fitted. Freezing matters. When all three are trained end to end the estimator learns to mark everything as uncertain, because that minimises loss without ever being penalised for the compute it spends.

Inference is a single forward pass for the majority of inputs. On GroundSum the expensive scorer fires on roughly one input in five, which is what makes ProbeCheck runnable on a single mid-range GPU at all. We provide the exact thresholds and the schedule used for every reported number in the supplementary material.

The cheap scorer is a two-layer projection over pooled encoder states. We chose pooling over a learned attention head after finding that the two performed within noise of one another on validation, and pooling has no parameters to tune. Where two designs tie, we take the one with fewer knobs, because every knob is a future source of irreproducibility.

3 Experiments

We evaluate on GroundSum, which contains 15,600 examples. We use the standard split released with the dataset and report faithfulness F1 on the held-out test portion. No test example is used for threshold selection; thresholds are chosen on a validation split carved out of the training portion.

Our primary baseline is NLIJudge, which we retrain under our own protocol rather than quoting published numbers. Retraining moves the NLIJudge figure by more than a point relative to the number reported by its authors, which is itself a small argument for retraining baselines rather than copying tables.

All runs use a single mid-range GPU. Every reported figure is the mean of five seeds. We report the mean rather than the best seed because the seed-to-seed spread on GroundSum is roughly one point of faithfulness F1, which is large enough that a best-of-five protocol would manufacture most of a claimed improvement.

We additionally run an ablation in which collapsing the probe to a sentence-level classifier is applied, holding every other setting fixed. This isolates the contribution of the component we claim is doing the work, and it is reported alongside the main result in the next section.

4 Results

ProbeCheck reaches 83.1 faithfulness F1 on GroundSum. The NLIJudge baseline reaches 76.4 faithfulness F1 under the identical protocol. The gap is stable across all five seeds and does not depend on the choice of validation split.

The ablation confirms where the gain comes from. With collapsing the probe to a sentence-level classifier, faithfulness F1 falls by 5.5 points, recovering most of the distance back to NLIJudge. We read this as evidence that the improvement is a property of the token-level attribution probe design rather than of the training schedule.

Cost is the other half of the claim. ProbeCheck completes the full GroundSum test pass on a single mid-range GPU, which the uncascaded configuration does not. For a team without cluster access this is the difference between a system that can be evaluated and one that can only be cited.

Breaking the GroundSum test set down by input length shows the gain is not uniform. The longest quartile of inputs accounts for most of the improvement, which is consistent with the token-level

attribution probe account: long inputs are where the cheap scorer is least reliable and where routing therefore has the most to decide.

5 Limitations

The most serious limitation of ProbeCheck is that the probe is trained on extractive summaries and its calibration drifts when the generator paraphrases heavily. We report this prominently because it determines whether the method is appropriate for a given deployment, and it is not visible from the headline faithfulness F1 figure.

A second limitation is evaluation scope. GroundSum is a single dataset in a single language, and we make no claim that the 5.5 point ablation gap would reproduce elsewhere. Cross-dataset transfer is the obvious next study and we have not run it.

References

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